16 There are some counters in a box.
Each counter is blue or green or red or yellow.

The total number of blue and green counters is twice the total number of red and yellow counters.

The number of green counters is $\frac{1}{6}$ of the number of blue counters.

Show that, to the nearest percent, the percentage of blue counters in the box is 57 %

Blue : green = 6 : 1 = 12 : 2

1 mark
Use B : G = 6 : 1.

B	G	R Y
12	2	7

1 mark
Match (B+G):(R+Y).

Blue + green = 14 parts.

Red + yellow = 7 parts.

Total = 12 + 2 + 7 = 21 parts.

1 mark
Blue ÷ all counters.

Blue percentage = 12 ÷ 21 × 100 = 57.142857... ≈ 57%

1 mark
Rounds to 57%.

(Total for Question 16 is 4 marks)